gr3_2@sys: ~/s5

```
gr3_2@sys:~/s5$ python3 json_client.py
gr3_2-JSON-RPC Client Demo
Request payload: {
  "jsonrpc": "2.0",
  "id": 1,
  "method": "evaluate",
  "params": [
    "2+3"
  ]
}
HTTP error: 400
evaluate(2+3) = None
gr3_2@sys:~/s5$
```

gr3_2@sys: ~/s5

```
 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
Last login: Wed Mar 18 09:21:38 2026 from 10.33.99.104
gr3_2@sys:~$ cd s5
gr3_2@sys:~/s5$
```

gr3_2@sys: ~/s5

```
 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
Last login: Wed Mar 18 09:21:50 2026 from 10.33.99.104
gr3_2@sys:~$ cd s5
gr3_2@sys:~/s5$ nano js_cl_rez4.py
gr3_2@sys:~/s5$ python3 js_cl_rez4.py
======================================
gr3_2  JSON-RPC Client (MathJS Hybrid)
======================================

--- PASUL 1: Apel de test (2+3) ---

[DEBUG] Request payload sent to server:
{
  "jsonrpc": "2.0",
  "id": 1,
  "method": "evaluate",
  "expr": "2+3"
}
[DEBUG] Response received from server:
{
  "result": "5",
  "error": null
}

RESULT evaluate(2+3) = 5

======================================
Introdu o expresie matematică (ex: 10*5 - 3): 10-5

--- PASUL 2: Trimitere expresie: '10-5' ---

[DEBUG] Request payload sent to server:
{
  "jsonrpc": "2.0",
  "id": 1,
  "method": "evaluate",
  "expr": "10-5"
}
[DEBUG] Response received from server:
{
  "result": "5",
  "error": null
}

--------------------------------------
 REZULTAT FINAL: 5
--------------------------------
gr3_2@sys:~/s5$
```

gr3_2@sys: ~/s5

```
   ---------------------------------------- 44.6/44.6 kB 861.1 kB/s eta 0:00:00
Installing collected packages: typing-extensions, setuptools, protobuf, grpcio, grpcio-tools
Successfully installed grpcio-1.78.0 grpcio-tools-1.78.0 protobuf-6.33.5 setuptools-82.0.1 ty
ping-extensions-4.15.0
(venv) gr3_2@sys:~/s5$ python3 -m grpc_tools.protoc -I. --python_out=. --grpc_python_out=. ca
lculator.proto
Could not make proto path relative: calculator.proto: No such file or directory
(venv) gr3_2@sys:~/s5$ nano calculator.proto
(venv) gr3_2@sys:~/s5$ python3 -m grpc_tools.protoc -I. --python_out=. --grpc_python_out=. ca
lculator.proto
(venv) gr3_2@sys:~/s5$ ls
calculator.proto   calculator_pb2_grpc.py  json_client.py
calculator_pb2.py  js_cl_rez4.py           venv
(venv) gr3_2@sys:~/s5$ nano grp_client.py
(venv) gr3_2@sys:~/s5$ nano grp_server.py
(venv) gr3_2@sys:~/s5$ nano grp_client.py
(venv) gr3_2@sys:~/s5$ nano grp_server.py
(venv) gr3_2@sys:~/s5$ nano grp_server.py
(venv) gr3_2@sys:~/s5$ python3 grp_server.py
[gr3_2-SERVER] gRPC Calculator server started on port 8702
[SERVER] Add called with a=2, b=3, result=5
[SERVER] Add called with a=2, b=3, result=5
[SERVER] Add called with a=5, b=7, result=12
[SERVER] Multiply called with a=5, b=4, result=20
```

gr3_2@sys: ~/s5

```
(venv) gr3_2@sys:~/s5$ python3 grp_client.py
[gr3_2-CLIENT] Connecting to gRPC server at localhost:8702
[CLIENT] Connected. You can now call Add, Multiply and Power.
Add(2, 3) = 5
Introduceti a pentru Add: a
Introduceti b pentru Add: 5
Traceback (most recent call last):
  File "/home/an3/gr3_2/s5/grp_client.py", line 108, in <module>
    main()
  File "/home/an3/gr3_2/s5/grp_client.py", line 77, in main
    a = int(a_str)
        ^^^^^^^^^^
ValueError: invalid literal for int() with base 10: 'a'
(venv) gr3_2@sys:~/s5$ python3 grp_client.py
[gr3_2-CLIENT] Connecting to gRPC server at localhost:8702
[CLIENT] Connected. You can now call Add, Multiply and Power.
Add(2, 3) = 5
Introduceti a pentru Add: 5
Introduceti b pentru Add: 7
Add(5, 7) = 12
Introduceti a pentru Multiply: 5
Introduceti b pentru Multiply: 4
Multiply(5, 4) = 20
Introduceti base pentru Power:
```

gr3_2@sys: ~/s5

```
(venv) gr3_2@sys:~/s5$ python3 grp_server.py
[gr3_2-SERVER] gRPC Calculator server started on port 8702
[SERVER] Add called with a=2, b=3, result=5
[SERVER] Add called with a=3, b=3, result=6
[SERVER] Multiply called with a=3, b=3, result=9
[SERVER] Power called with base=3, exponent=3, result=27
```

gr3_2@sys: ~/s5

```
gr3_2@sys:~/s5$ netstat -a -n -p -t | grep 8702
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
tcp6       0      0 :::8702                 :::*                    LISTEN      1958586/pytho
n3
gr3_2@sys:~/s5$ netstat -a -n -p -t | grep 8702
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
tcp6       0      0 :::8702                 :::*                    LISTEN      1958586/pytho
n3
gr3_2@sys:~/s5$
```

gr3_2@sys: ~/s5

```
(venv) gr3_2@sys:~/s5$ python3 grp_client.py
[gr3_2-CLIENT] Connecting to gRPC server at localhost:8702
[CLIENT] Connected. You can now call Add, Multiply and Power.
Add(2, 3) = 5
Introduceti a pentru Add: 3
Introduceti b pentru Add: 3
Add(3, 3) = 6
Introduceti a pentru Multiply: 3
Introduceti b pentru Multiply: 3
Multiply(3, 3) = 9
Introduceti base pentru Power: 3
Introduceti exponent pentru Power: 3
Power(3, 3) = 27
[CLIENT] Finished. Results logged to grpc_client_log.txt
(venv) gr3_2@sys:~/s5$
```

gr3_2@sys: ~/s5

```
gr3_2@sys:~/s5$
```